

Anomaly detection for sustainable automotive manufacturing

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Abstract: The automotive sector provides society with the means to move people and goods, however the increasing need for climate change mitigation places responsibility on automotive manufacturers to develop low-carbon technologies to meet these needs. These rapid technological developments are prone to high costs, skill gaps, and sub-optimal energy-intensive practices, particularly within the testing environments for automotive components. Automated safety thresholds are in place to detect large fluctuations in real-time data, whilst the detection of other, more nuanced faults is manual, but these have the risk of undermining vehicle development. As such, this process is resource intensive, such as the personnel time and energy consumed to rerun tests containing faults, and the financial costs associated with these. Here we show the ability of unsupervised, data-driven machine learning based anomaly detection methods to identify anomalous time-periods within an automotive test, including faults in either the test component or in the test facility equipment, without the need for training data. We compare the performance of three clustering algorithms – *k*-means, agglomerative clustering, and DBSCAN – based on their run time and ability to create defined anomalous clusters of the two anomalies. *K*-means was able to identify the two anomalies with eight total clusters in half the time of agglomerative clustering. DBSCAN clustered the data in half the time of *k*-means however was unable to create defined anomalous clusters. These results illustrate the potential for unsupervised data-driven anomaly detection to operate within automotive manufacturer testing environments. These methods provide a low-cost digital solution to the resource demands associated with the traditional processes used by automotive manufacturers when developing sustainable transport options.

1. Introduction

In 2014, the manufacturing sector was responsible for 20% of global carbon dioxide (CO₂) emissions [1]. A further 20% was produced by the transport sector [2], which rose to 23% of global CO₂ emissions in 2022 [3]. This trend illustrates the detrimental impact that automotive industries continue to have towards climate change. Automotive manufacturers have responded to calls for decarbonisation of the sector, encouraged by political actors and societal shifts,

by exploring low-carbon technologies and streamlining operations to improve efficiency and reduce time-to-market of these options [4], [5].

Automotive manufacturing combines several processes, each with its own opportunities and challenges for automation and increased sustainability [6]. The automotive development phase is an iterative process involving product testing and prototype development, with the data generated from testing being used for component design decisions (Figure 1).

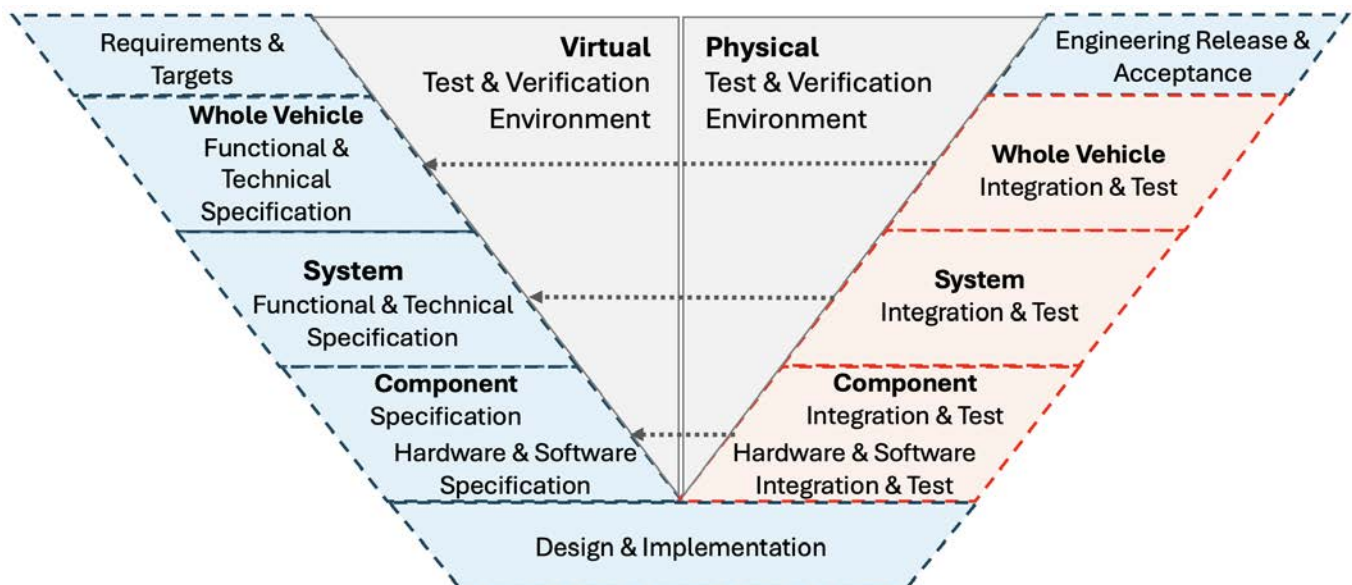


Figure 1. Automotive development cycle V Model. Sections shaded in red indicate physical testing of vehicle systems where anomaly detection can be applied.

Automotive development is strongly reliant on physical testing because the ability to virtually model the behaviour of components and systems is not adequate to reach the sign-off requirements [7], [8]. Physical testing is resource intensive due to the type of physical prototypes that are created and analysed. Batteries for example require long testing cycles to characterise their aging and degradation. Anomalies in the testing caused by either the unit under test (prototype) or in the test equipment confound this by causing redundant data acquisition requiring re-running of tests. Depending on how quickly such an anomaly is detected, the impact can become significant.

Identifying anomalies is a significant challenge during the test phase as thousands of variables are typically recorded during each test, and data quality diagnostics are predominantly based on past experience of the engineering team which may not be fully applicable as new low-carbon technology options are developed [9], [10].

In some instances, insufficient detection of anomalies in the test phase can lead to approval of a component design which informs other component or system design decisions further up the V-model (Figure 1). This can result in delayed product launches, compromised product performance and safety, and increased cost and energy consumption due to longer development times [11]–[13]. If these decisions based on low-quality data propagate all the way through the development process to product release and acceptance, they can result in severe financial and reputational consequences for the manufacturer if product recalls are required.

Given the challenge and potentially large impacts associated with obtaining high-quality test data, manufacturers aim to (1) minimise the anomalies occurring in their tests, and to identify these as soon as possible, and (2) shift towards virtual testing rather than physical testing to reduce the time and energy invested into producing prototypes. However, virtual models require high quality data from physical testing, therefore this research focuses primarily on the first aim in order to facilitate the second.

Anomaly detection (AD) is a machine-learning approach to identify unusual, unexpected, or previously unseen patterns in data, known as anomalies [14], [15]. There are low upfront costs associated with AD and manufacturers can receive benefits from its implementation. These include:

- (1) reducing operational costs from ensuring first-time-right testing and creating high quality datasets to feed into virtual models;
- (2) increasing engineers' trust in data generated, leading to reduced stress associated with decision making;
- (3) autonomous AD reduces the need for a domain expert to monitor the test bed, thus mitigating skills gaps and freeing up time for competing priorities, and
- (4) minimising wasted energy consumption from physical tests running when the data quality is compromised.

These benefits to the manufacturer can propagate into global benefits in line with the United Nations 17 Sustainable Development Goals (SDGs)[16]. For example, low-carbon transportation options (SDG 11) which support global goals to reach net zero CO₂ emissions (SDG 13) could be provided to individuals at lower costs (SDG 7) if development times and associated resource consumption are reduced (SDG 12). Additionally, if the development of low-carbon transportation technology is twinned with organisational and energy efficiency efforts, such as deploying AD, the energy demand

and embedded CO₂ associated with the vehicle development cycle may be reduced.

Given the multi-faceted benefits that AD can provide, this paper applies three data-driven, unsupervised clustering approaches to detect anomalies in test data typical of that collected during the development of a diesel-powered passenger vehicle. Unsupervised machine-learning is selected to remove the need for labelled training data which is an impractical expectation in commercial environments due to resource constraints [15]. Clustering algorithms specifically increase the potential for operability of AD by also removing the need for any training data, such as is required for model-based unsupervised machine-learning approaches, such as autoencoders [17]. This is especially beneficial for the automotive testing and development industry when working on new technologies which have no pre-existing or accessible data to train a model. The approaches presented in this paper therefore offer the opportunity to detect anomalies on new technologies or pieces of test equipment during a test run and without the need for multiple test repeats or to complete a full test run.

Details of the dataset used are described in section 2 and section 3 outlines the clustering methodologies applied for AD. Results are presented in section 4, before their significance for the sector and sustainability are explored in section 5. Conclusions of the work completed are presented in section 6, with areas of future work also suggested.

2. Data Description

The dataset evaluated within this study was generated by [18], where a chassis dynamometer was used to test a full vehicle during typical vehicle development duty cycles (NEDC and WLTC cycles). These tests last twenty to thirty minutes and are commonly used by development engineers and legislative bodies to evaluate vehicle performance metrics including fuel consumption and emissions. Several sub-systems (such as fuel supply, after treatment, electrical system, etc.) interact during a full vehicle test, thus a large number of sensors are installed to monitor their behaviour. In this study, the dataset comprised of approximately 200 variables, with a combination of measurements from sensors, calculated quantities derived from combinations of raw measurements (such as absolute humidity and mass flow of emissions species), and state variables recorded from the vehicle's electronic control unit. The dataset also includes quantities relating to the test facility performance, including the robot driver behaviour.

The purpose of the test campaign was to demonstrate stable and repeatable vehicle operation over a series of drive cycle tests. These tests were performed over several days (<10), including thermal conditioning of the vehicle between tests. Following data acquisition, manual data processing occurred over two weeks at which point unwanted variation between tests was identified. Four anomalies were identified by [18] using expert-led analysis. The authors proposed Partial Least Squares (PLS) Regression to automatically identify the anomalies. Both the expert-led and PLS approaches required 4-8 test repeats before an anomaly was detected [18].

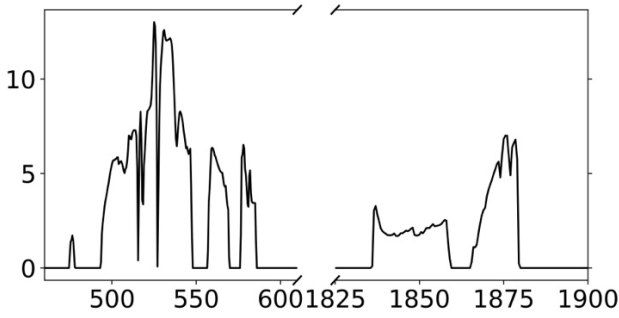


Figure 2: Anomalous behaviour of the Post-Fuel Injection variable within one test. Normal behaviour would show this variable to remain at zero.

One anomaly was selected for this study to evaluate the performance of clustering algorithms to detect the anomaly, when presented with only two test cycles (Figure 2). The anomaly related to the behaviour of the vehicle controller using an exhaust heating strategy. This varying behaviour was likely due to small differences in the thermal state of the vehicle during the duty cycle, but manifested itself through additional fuel injected into the exhaust system. The anomaly can be identified through the throttling of intake air flow, fuel injector activity, and increased exhaust gas temperature. The latter two characteristics were isolated for this study, resulting in a four variable dataset including fuel injector activity and three temperature measurements along the exhaust pipe.

Three variants of the dataset were created prior to clustering to test the effect of data pre-processing methods (Figure 3). This was conducted due to the different scales of the automotive variable data being expected to impact clustering performance. One dataset was left in its raw form, one was standardised such that all values were scaled to between 0 and 1, and one was normalised so that the mean and standard deviation of the data were 0 and 1 respectively. Comparison of the algorithms performance across the three pre-processed datasets is presented and discussed in sections 4 and 5 below.

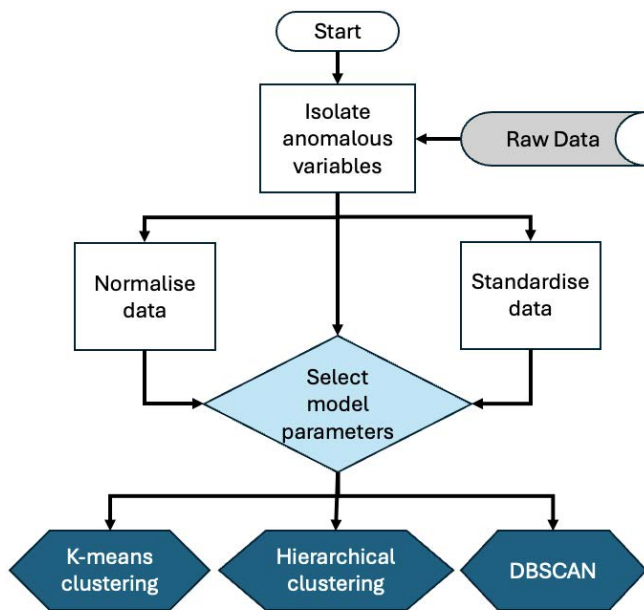


Figure 3. Data processing methodology to prepare secondary data for cluster analysis.

3. Methodology

This section describes three clustering algorithms, their hyperparameter selection process, and performance evaluation methods. Algorithms were selected based on their ability to perform AD without training data, their reported performance in comparative literature, and due to representing three strategies for clustering - centroid, agglomerative, and density-based [10], [11], [19], [20].

3.1 Algorithm descriptions

The k -means algorithm is an iterative method for clustering a dataset into k distinct groups [21]. By selecting an initial k cluster centres, all data points are assigned to their closest centre using Euclidean distance or Manhattan distance. The cluster centres are redefined to be the average of the points contained within the cluster and data points are then re-allocated to the new nearest adjacent centre [22]. This re-assignment of data points to cluster centres is repeated until all points remain in the same cluster.

Agglomerative clustering implements a hierarchical model to form clusters. The agglomerative form creates clusters by merging pairs of data, starting with the two closest data points, until all data are combined into a single cluster. The distance between data points is calculated using Euclidean or Manhattan distance. A linkage criterion determines how the algorithm merges clusters; an average linkage merges clusters based on the average distance between them, complete linkage uses the maximum distance between clusters, and single linkage uses the minimum distance clusters [23]. Both distance metrics and all three linkage criterion were evaluated in this study.

Density-based Spatial Clustering of Applications with Noise (DBSCAN) is the most common density-based clustering algorithm within literature [21]. It forms high-density clusters of an arbitrary shape separated by areas of low-density [24].

3.2 Hyper-parameter selection

k -means and agglomerative clustering require the number of clusters, k , hyperparameter to be defined. Appropriate selection is important to achieve clusters which are dense and well separated. Both the Davies-Bouldin Index (DBI) and the Calinski-Harabasz Index (CHI) were used to select k in this study.

Both metrics calculate the average similarity of each cluster. DBI calculates similarity as the ratio of within-cluster distances to between-cluster distances [25]. Whilst CHI is the ratio of between-cluster distances to within-cluster distances [26]. A low DBI or high CHI value is desirable to ensure compact, well-defined clusters. Given this inverse relationship, the two metrics can be calculated, scaled between naught and one, and plotted together; an optimum k value can be identified where the lines intersect. k values between 2 and 20 were explored for suitability.

The number of clusters, k , is not defined for DBSCAN, instead a threshold for the minimum number of data points, $minPts$, to form a cluster with a radius eps (ϵ) are selected. Data points that do not fit within these criteria are categorised as noise. The minimum number of data points within a cluster can be defined as the number of dimensions within the data multiplied by two [27]. For the dataset in this study containing 4 variables, $minPts$ was set at 8. The cluster radius, eps , was selected by calculating the average distance between

each point and its k -nearest-neighbours, where k is the minimum number of data points. Plotting the average k -distances allows eps to be chosen as the point of maximum curvature, or elbow point [20], [27].

3.3 Evaluation

The outputs of the clustering algorithms were evaluated in two ways. A subjective, qualitative evaluation of performance – that is, the ability of the algorithms to create distinct clusters which only contain the anomalous data points – was the primary focus of this study.

Results were considered poor if both anomaly spikes were represented by clusters containing anomalous and normal data. Moderate performance was achieved if one anomaly spike was represented by at least one cluster containing only anomalous data, but the second anomaly spike was represented by clusters containing anomalous and normal data. Performance was considered good if at least one cluster contained only anomalous data and was present in both anomaly spikes.

The second evaluation technique was a quantitative method to assess operability by recording the time taken for each algorithm to cluster the dataset. For this, the Python package *time* was used [28]. The same hardware setup - a Macbook Pro (2020), 16GB RAM - was used for all algorithm trials.

4. Results

The results of this study are presented by algorithm in this section before being discussed comparatively and with regards to operability in section 5. Each subsection highlights the performance of an algorithm, including the number and distribution of clusters, time taken, and evaluated based on the methodology described in the previous section.

4.1 k -means

k -means was able to create distinct clusters for two values of k - eight clusters and eleven clusters – when applied to the scaled (Figure 4) and normalised (Figure 5) dataset respectively. This algorithm performed poorly – no distinct anomalous cluster was created - for all k values on the raw dataset. This result aligns with the original assumption that led to the creation of the scaled and normalised datasets, whereby the variability of the raw data would be incompatible with k -means clustering.

The performance of k -means on the scaled and normalised datasets when k was set to eight and eleven respectively was evaluated as good, such that distinct, independent clusters were formed on the anomalous data spikes. With eight clusters on the scaled dataset, k -means produced one cluster (shown in purple) which was independent to the anomalous data spikes (Figure 4). Three clusters primarily contained normal data but also included anomalous data points – true negatives – and four distinct clusters were made up of only normal data. Similarly, when k -means was applied to the normalised dataset with eleven clusters (Figure 5), one cluster (shown in brown) was distinct to the anomalous data spikes. Of the remaining ten clusters, four contained a combination of normal and anomalous data points, and six contained only normal data.

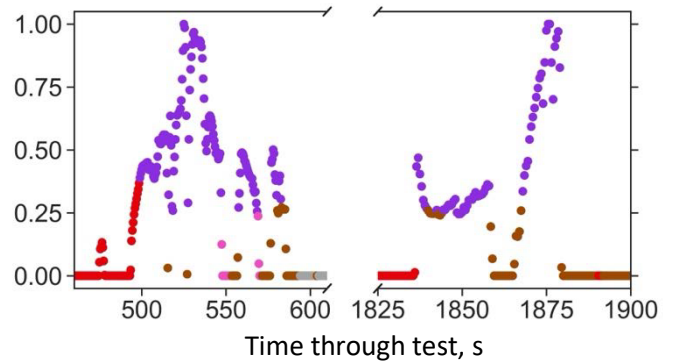


Figure 4: Results of k -means clustering on the scaled dataset, with $k=8$ clusters. One distinct anomalous cluster shown in purple.

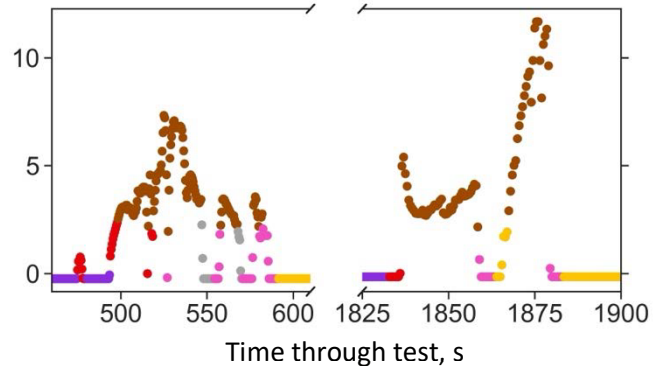


Figure 5: Results of k -means clustering on the normalised dataset, with $k=11$ clusters. One distinct anomalous cluster shown in brown.

k -means' runtime on scaled dataset where $k=8$ and normalised dataset where $k=11$ ranged between 74-76ms and 92-94ms respectively. Given a 20% faster completion on scaled dataset and with fewer clusters, scaled data appears preferential for the automotive application.

4.2 Agglomerative Clustering

Eighteen permutations were tested for agglomerative clustering, and the performance of each is summarised in Table 1. The algorithm was unable to differentiate between the anomalous data spikes and other data points within the Raw dataset, regardless of linkage criterion or distance metric, thus all permutations were considered poor. Clustering with a single linkage and Manhattan distance resulted in one anomalous spike being detected when k was set to twelve but not both anomalous spikes, thus its performance is considered moderate. Good clustering performance was achieved using the scaled data with average linkage and both distance metrics, such that both anomalous data spikes were represented by at least one cluster containing only anomalous data points.

Table 1: Evaluation of 18 permutations of agglomerative clustering using raw, scaled, and normalised datasets. Iterations vary the distance metric and linkage criterion

Linkage	Distance	Dataset type		
		Raw	Scaled	Normalised
Single	Euclidean	Poor	Poor	Poor
	Manhattan	Poor	Moderate	Poor
Average	Euclidean	Poor	Good	Poor
	Manhattan	Poor	Good	Poor
Complete	Euclidean	Poor	Poor	Poor
	Manhattan	Poor	Poor	Poor

Agglomerative clustering had the longest and most variable runtime of the three algorithms tested, taking between 156-175ms to cluster the scaled dataset with average linkage using the Manhattan distance, and 167-174ms for the Euclidean distance.

When clustered using the Euclidean distance with k set to eight, four distinct clusters across both anomalous data spikes were achieved (Figure 6). Only one cluster was created containing only normal data points and the remaining three clusters contained predominantly normal data with some anomalous values. In comparison, k was set to nine when the Manhattan distance was used to cluster the scaled dataset with average linkage (Figure 7). This permutation also resulted in four distinct anomalous clusters to be formed, however three clusters contained only normal points and the remaining two clusters had normal and anomalous data points.

From these results, Manhattan distance may have a greater ability to distinguish between the anomalous and normal data due to creating a greater number of clusters containing only normal data points compared to when using the Euclidean distance. However, given that both distance metrics were able to detect the two anomalous data spikes in the scaled dataset with an average linkage, they appear less influential on the performance on agglomerative clustering for these data. In contrast, the linkage criterion chosen was highly influential as defined clusters were created using an average linkage, but not single or complete linkages.

These results align with those from the k -means algorithm as the average linkage performs in a comparable way to k -means, and both found eight clusters to successfully cluster the anomalous data within the scaled dataset.

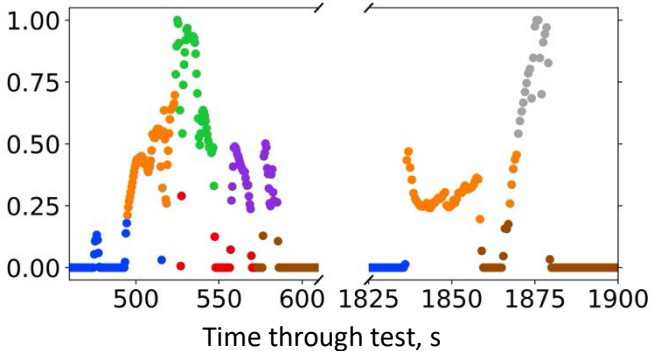


Figure 6: Agglomerative clustering with average linkage and Euclidean distance on scaled data, with $k=8$. 4 distinct anomalous clusters: orange, green, purple, and grey.

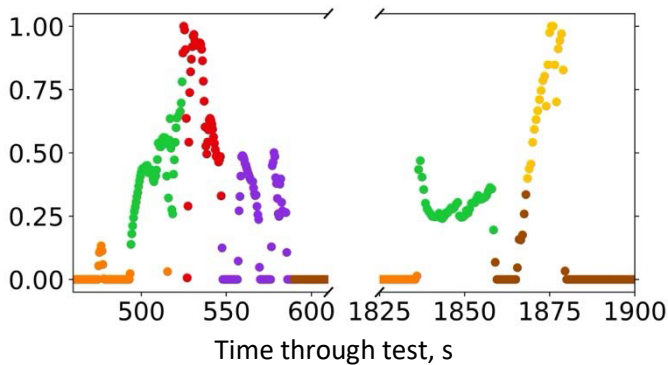


Figure 7: Agglomerative clustering with average linkage and Manhattan distance on scaled data, where $k=9$. Three distinct anomalous clusters: green, red, and yellow.

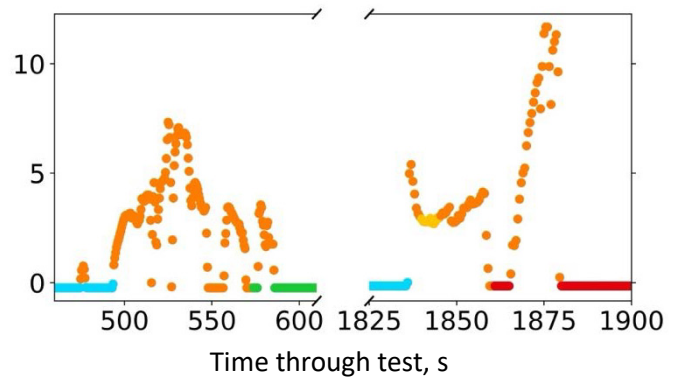


Figure 8: DBSCAN results with the normalised dataset. 20 clusters were produced using $eps=0.18$ and $minPts=8$. One distinct anomalous cluster shown in yellow.

4.3 DBSCAN

DBSCAN had the fastest runtime of the three algorithms of 45-47ms, however it typically had poor performance when clustering the automotive datasets. It was unable to produce clusters independent to the anomalous data spikes across all three datasets, despite being the only algorithm to cluster almost all of the anomalous data points into the same cluster. Figure 8 shows DBSCAN's performance on the normalised dataset, where the anomaly spikes can be seen to be represented almost exclusively by one cluster (orange data points), with a few points assigned to two additional clusters (yellow and blue). Twenty clusters were produced in total, with three containing anomalous data points whilst seventeen contained only normal data points. DBSCAN produced similar results using the scaled dataset.

5. Discussion

The results show the ability of unsupervised, data-driven clustering algorithms to effectively detect anomalies with only two drive cycle tests. This performance is superior to previous work on these data as clustering AD was able to detect the anomalies three-times faster than expert-led and PLS analysis, which required six tests [18]. In practise, AD would have prevented at least four poor quality test runs operated over four days, and two weeks of post-processing and analysis that were required before anomalies were identified. Preliminary results indicate that these methods are also applicable to other automotive test environment datasets, such as for durability testing of electric powertrains. This is further supported by [29] and [30] successfully applying k -means clustering with Euclidean distance to detect vibration anomalies within jet engines and traffic flow data anomalies respectively. Both cluster a reduced dataset by using a priori knowledge to extract features of interest, but [30] found k -means to be scalable when applied to generated and real data.

These findings show promise for AD to reduce wasted resources associated with automotive testing, such as time, energy consumption, and the financial costs associated with each, thus bolstering overall sustainability of automotive manufacturing.

When compared across the two evaluation criteria of this study – clustering performance and run time – k -means clustering is most suitable on this automotive testing dataset. The clustering performance of k -means is similar to agglomerative clustering, by being able to produce clusters

which exclusively contained anomalous data points. However, k -means presented greater distinction between anomalous and normal data by having more clusters which were exclusively normal data, rather than clusters which contained both normal and anomalous data points.

[11] and [21] also found hierarchical clustering and k -means to outperform other clustering algorithms when applied to steering wheel angle sensor anomalies and product pricing anomalies respectively. [21] compared both to seven other algorithms including DBSCAN, where DBSCAN performed third-to-last based on the Borda count method due to creating several single-data point clusters. This behaviour corresponds to our results where DBSCAN created twenty clusters compared to approximately ten created by k -means and agglomerative clustering.

Depending on how clusters are labelled and interpreted, mixed clusters may be viewed as containing either false positives – normal data perceived as anomalous – or true negatives – anomalous data perceived as normal. Within the automotive development process false positives are typically more favourable than true negatives in order to ensure any product defects are discovered and to aid understanding of a new technology's behaviour. However, careful selection of this balance between false positives and true negatives will be vital during implementation of an AD methodology to maximise the ability for an engineer to appropriately design a component versus minimising the interference and annoyance caused by an algorithm identifying too many false positives.

k -means was also superior to agglomerative clustering regarding run time, producing a similar number of clusters up to 2.2 times faster than agglomerative clustering. DBSCAN had a notably faster run time than both of these algorithms, being up to 3.8 and 2.0 times faster than agglomerative clustering and k -means clustering respectively. However, despite DBSCAN producing clusters twice as fast as k -means, it was unable to produce distinct anomalous clusters. Additionally, the requirement for more expert input to optimise hyperparameters reduces its applicability within a commercial manufacturing setting. Interestingly, this contrasts findings by [31] who identify a flaw of DBSCAN as being highly demanding of computational resources.

Defining the balance between run time and AD performance to select an algorithm may vary based on application within the automotive test and development manufacturing process. Component tests running for a short time, such as less than one hour, may benefit from better clustering performance to identify anomalies in relatively small datasets. In contrast, engineers working on product tests lasting for more than twelve months may prefer an algorithm which provides results faster, even if more false positives are detected. The integration of the algorithm will be a key part; for example choosing whether AD should run during the test to detect anomalies near-real time, or if a test will conclude prior to checking the data for anomalies. This latter example is comparable to this study, where anomalies were detected on completed test data. Crucially, the unsupervised, data-driven machine learning approaches explored in this work were able to detect both anomalies with only two test runs, compared to the original study conducted which required 4-8 test runs to detect the anomalies [18].

5.1 Limitations and Future Work

Limitations of this work revolve predominantly around subjectivity associated with hyperparameter selection and clustering performance evaluation. Whilst the DBI and CHI were utilised to minimise subjectivity and increase reproducibility, the selection of the k -means and agglomerative clustering k hyperparameter is largely based on qualitative interpretation of the DBI-CHI plot and therefore can be prone to ambiguity. For example, the 18 iterations explored for agglomerative clustering were chosen based on DBI-CHI plots however most of the clustering results were poor. This hyperparameter selection therefore poses a challenge in the operationality of these algorithms. Subjectivity is also a limitation of these approaches during the performance evaluation. This study was able to define poor, moderate, and good performance due to the anomaly being known, however in operational settings an engineer will need to evaluate the clusters produced without prior knowledge. Future work should look to create a quantitative analytical method to determine anomalous clusters to maximise reproducibility and ease of application to industry.

6. Conclusion

This work demonstrates the ability of unsupervised clustering algorithms to detect anomalies within time-series data from an automotive development cycle, without the need for training data. Anomalies were detected after only two drive cycle tests, greatly improving previous AD attempts on these data which required six repeat tests [18]. If a clustering AD algorithm had been deployed on the original test campaign, operational efficiency could have been improved by over 66%. This is achieved by avoiding poor-quality drive cycle tests, reducing the resources dedicated to manual post-processing, and reducing the time and energy demands associated with both.

k -means clustering performed best, by detecting the two anomalous data spikes with few, distinct clusters and with a run-time 2.2x faster than agglomerative clustering. Whilst DBSCAN generates clusters 2x faster than k -means on this dataset, it's high sensitivity to hyperparameter selection and inability to create distinct anomalous data clusters suggest that it is less compatible with commercial AD applications in the automotive manufacturing sector.

Given the capabilities of AD on time-series automotive development data, it can support the automotive manufacturing sector in improving operational efficiency, reducing wasted energy consumption associated with low-quality test runs, and reducing the time-to-market for low-carbon transport technologies to reach the public domain. As a result, AD can reduce costs associated with automotive testing by:

- (1) removing the need for training data, thus saving money on model training and increasing accessibility, especially to companies with low computational capabilities;
- (2) reducing the delayed costs associated with making design decisions on bad data;
- (3) reducing the number of tests required; and
- (4) reducing wasted effort from employees.

Together, these impacts will allow the automotive manufacturing sector to improve its sustainability and contribute to global climate change mitigation efforts.

7. Acknowledgments

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